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Permo–Triassic paleomagnetism of the North Pamir: tectonic implications

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Abstract

After stepwise thermal demagnetization of Upper Permian tuffaceous red beds (localities 1 and 2) and Triassic volcanics (locality 3) of North Pamir (Central Asia), the following mean directions of characteristic magnetization components were obtained: (1) Upper Permian (locality 1): D = 179.3°, I = −38.0°, α95 = 5.4° (29 samples); (2) Upper Permian (locality 2): D = 137.4°, I = −42.0°, α95 = 6.6° (20 samples); (3) Middle–Upper Triassic (locality 3): D = 226.8°, I = −63.8°, α95 = 4.4° (16 samples).

The conglomerate and fold tests are positive for the first and third results, whereas no field test could be performed for the second one. The Late Permian locality mean inclinations are statistically indistinguishable, and the observed difference in mean declinations closely matches that in structural trends between localities 1 and 2. This constitutes a positive fold test and, at the same time, points to oroclinal bending of the Pamir structures, thus confirming an earlier conclusion about the secondary origin of this arc. Permian and Triassic paleolatitudes, together with earlier published Early Cretaceous data, fall close to the European reference inclination curve. This implies that North Pamir has been close to the Eurasia landmass at least since the Late Permian, and the major part of the India–Eurasia convergence was absorbed to the south of this zone.

Keywords: Pamirs; paleomagnetism; Permian; Triassic; plate tectonics

1. Introduction

The part of the Alpine belt just to the north of India and between the Tibet, in the east, and Afghanistan, in the west, is known as the Pamir–Punjab syntaxis. The northern part of this is occupied by numerous ranges and plateaux known as the Pamir (Fig. 1). This area was strongly affected by the India–Eurasia collision [1–3]. Also clear is the fact that the pre-collisional structure of the Pamir resulted from considerable horizontal movement [4]. For instance, the spatial distribution of Asselian fusulines implies that, in the Early Permian, the North Pamir was at low northern latitudes, whereas the South Pamir was at mid-latitudes of the southern hemisphere [5]. The available data, however, are insufficient for establishing the magnitude of horizontal movements.

Paleomagnetic data could be of utmost importance here. Early Cretaceous results from the Tadjik basin and along the Pamir rim [6–9] have revealed inclinations quite close to those expected from the

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Eurasian apparent polar wander path, APWP, thus implying close proximity of this area to extra-Alpine Eurasia. Also, it has been shown that the curved form of the External Pamir zone resulted from oroclinal bending [9]. From geometrical considerations and rotations deduced from paleomagnetic data, the northward translation of this zone must have been about 300 km [9].

Older paleomagnetic data from the Pamir are scarce. Data on Lower Permian limestones from the South Darvaz Range from the North Pamir zone [10] are hardly reliable, because they are based only on blanket cleaning at 200°C, although the fold test is positive. Another result from the same area has not yet been published in detail and was just referred to by Klishevich et al. [11]. No data are known from the Central Pamir. Data on Permian rocks from southeast Pamir implies its position at low northern latitudes [12]; however, the result is untrustworthy, being mainly based on the time-cleaning technique (storage test); the data on a few samples subjected to blanket cleaning at 400°C do not pass the reversal test. One of the sampling sites according to A.N. Khramov (personal communication), was also studied by this author. Most samples revealed only a single component whose direction is close to the present day field and the mean direction from [12] (Bazhenov, unpublished data). The results imply a severe recent remagnetization of Permian rocks in the South Pamir zone, in accord with intense Alpine deformation, as deduced from geological data [4].

Below are presented new paleomagnetic data on Permian and Triassic rocks from the North Pamir. This study is part of a project directed at the paleo-

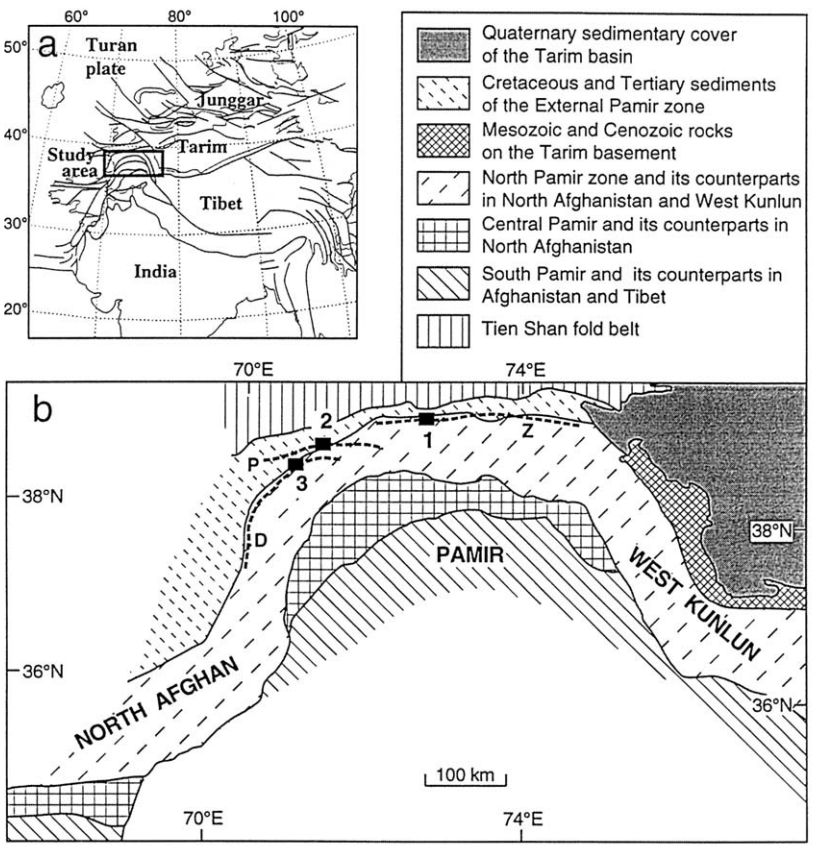


Fig. 1. (a) General location map showing major faults and areas with elevations greater than 2 km (shaded) simplified after Cobbold and Davy [2]. (b) Tectonic zonation of the Pamir simplified after Burtman and Molnar [14]. Sampling localities shown as solid rectangles are numbered as in the text. Thick dashed lines labelled D, P and Z are for the Darvaz, Peter the First and Zaalay ranges, respectively.

magnetic study of the Permo–Triassic of Central Asia and the decipherment of the tectonic evolution of this region.

2. Geological setting and sampling

The pamir can be divided into four zones; from the north to south: the External, North Pamir, Central Pamir and South Pamir zones [4]. These zones can be traced both westward and eastward (Fig. 1b) and, in turn, are subdivided into a number of tectonic units. The following description will be limited to the first two main zones.

The External zone comprises Mesozoic to Cenozoic sedimentary rocks and is regarded as a strongly deformed band between the Tadjik basin, in the west, and the western part of Tarim, in the east. No angular unconformities are reported from the Mesozoic–Paleogene part of the section. The first folding locally occurred in the Early Miocene [13], and the main deformation took place in the Late Miocene to Quaternary. This deformation is connected with the India–Eurasia collision and indentation of the rigid Pamir block into more ductile adjacent areas [1–3].

The North Pamir zone, mainly comprising Carboniferous–Permian rocks, can be traced westward into the western Hindu Kush and Parapamiz in Afghanistan and eastward into the western Kunlun in China [14,15] The southern boundary of the North Pamir zone coincides with the regional Tanymas–Akbaytal Thrust. Intense Triassic–Early Jurassic volcanism is reported from the western (Parapamiz) part of the belt [15]; volcanics of the same age are also locally present in North Pamir [16].

The above two zones have a similar curved outline. The boundary between them is drawn along the Darvaz–Karakul Fault [17], separating Mesozoic–Cenozoic rocks of the External zone from Paleozoic rocks of the North Pamir zone. Stratigraphic contacts, however, were locally found between Mesozoic and Paleozoic rocks [18], and the Darvaz–Karakul Fault is probably a younger feature.

At locality 1 in the Zaalay Range (Fig. 1b), the Permian sequence, separated by a fault from Lower Carboniferous volcanics, comprises Lower Permian flysch and about 500 m thick light-coloured limestones, which are conformably overlain by violet and green sandstones and siltstones with some limestone layers (Gundara Formation). The counterparts of this formation to the west are of the early Late (Ufimian, c. 265 Ma) Permian age [19]. The Gundara rocks are overlain with erosional and probably weak angular unconformity by Late Permian conglomerates. With erosional disconformity, these conglomerates are overlain by a more than 2,000 m thick sequence of clastics, lava and tuffs of the Late Triassic–Early Jurassic Minteke Formation, metamorphosed to greenstone facies. Triassic rocks are separated from the Lower Cretaceous by faults, and it is unclear whether any deformation had taken place before the Early Cretaceous. Hand samples were taken from both violet (34 samples) and green (10 samples) varieties of the Gundara sandstones at six sites (K1–K6), spanning the entire section of the formation. The beds are rather uniformly dipping north to northeast 20–50°. Twenty boulders of various lithology were sampled from the overlying Upper Permian conglomerates (site Y1); many boulders are visibly similar to the Gundara sandstones. Triassic volcanosedimentary rocks were sampled at five sites (54 samples) from the lower half of the formation from beds of various attitude.

At locality 2, on the southern slope of the Peter the First Range (Fig. 1b), the Permian section is similar to that at the previous locality. Violet-brown, tuffaceous, coarse- to medium-grained sandstones of the Gundara Formation were sampled from site S1, whereas mostly fine-grain sandstones and siltstones were taken at site S2. Bedding attitudes at both sites are similar; unfortunately, other outcrops are at inaccessible heights.

At locality 3, an about 2,000 m thick pile of volcanics and coal-bearing terrigenous rocks is separated by the Darvaz–Karakul Fault from the Upper Permian on the northern slope of the Darvaz Range (Fig. 1b). The Middle–Late Triassic (but pre-Rhaetian) age of the volcanics is based on paleoflora [18]. The volcanics are disconformably overlain by Liassic–Upper Cretaceous sedimentary rocks. No angular unconformities within the Mesozoic were reported from this area. The main folding here occurred in the Late Cenozoic, with a maximum in the Pliocene. In contrast to locality 1, Triassic rocks are not metamorphosed and show no signs of penetrative deformation, such as schistosity or cleavage. Paleo-

magnetic samples were taken from the upper member of the Triassic section, consisting of numerous flows of alkali basalts and andesites and conglomerates with lava boulders. At site T1, 30 hand samples were taken from at least 13 vertically dipping flows; 18 lava boulders were sampled from conglomerates; 13 samples from at least 8 volcanic units were taken from site T2, with the beds in overturned position. Lower Cretaceous red beds from the same locality have also been studied [9].

At each sampling site, samples were oriented with a magnetic compass and spaced several meters across the sections over a true thickness of up to several hundred meters.

3. Methods

One to three, usually two duplicate specimens from each hand sample were subjected to stepwise thermal demagnetization in a home-made oven with a residual field of about 10 nT. Two specimens from most boulders were also treated, in order to estimate the within-boulder scatter of paleomagnetic directions and exclude the possibility that this scatter could affect the outcome of the conglomerate test. All measurements of natural remanent magnetization, NRM, were made on a Czechoslovakian spin magnetometer JR-4 in Moscow, except for a few samples which were a.f. demagnetized and measured

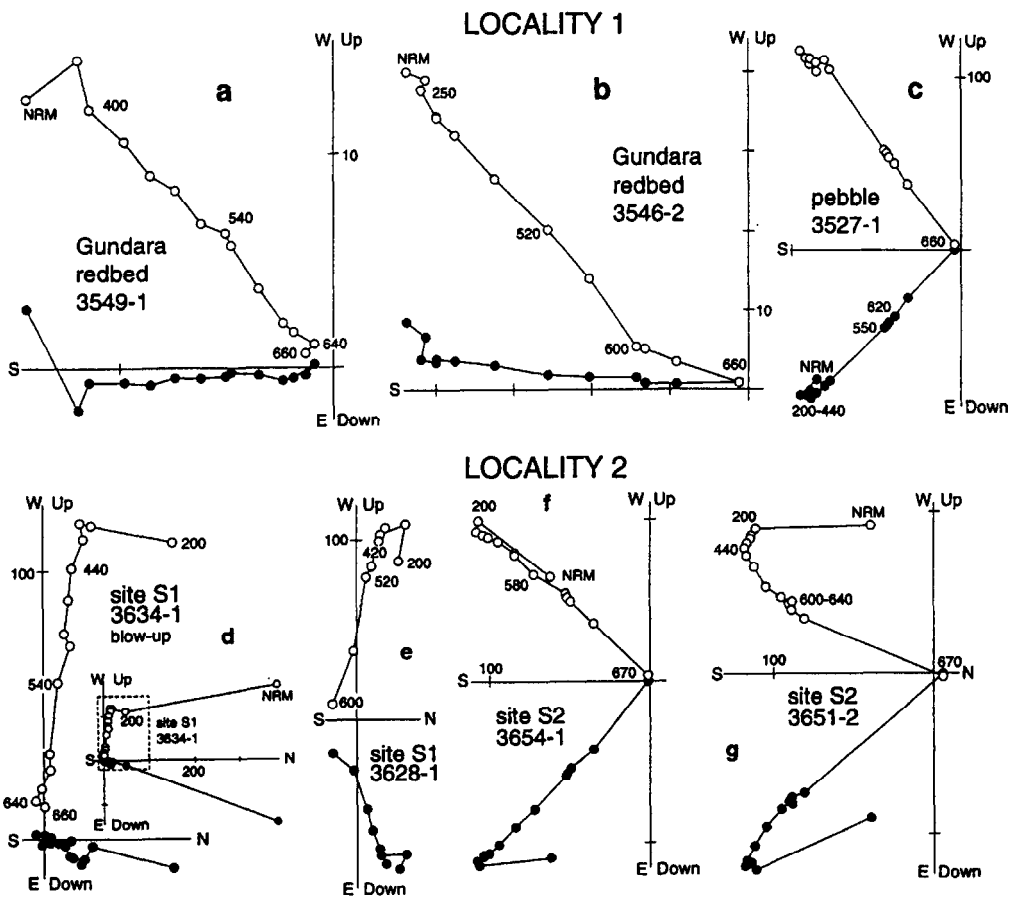


Fig. 2. Representative thermal demagnetization plots of samples from the Upper Permian Gundara Formation (a,b and d–g) and Upper Permian conglomerates (c). Data are plotted in stratigraphic coordinates. ● = vector endpoints projected onto the horizontal plane; ○ = vector endpoints projected onto the vertical plane. Steps are in degrees Celsius. Magnetization intensities are in mA/m.

with a 2G cryogenic magnetometer in Paris. Isolated components [20] from each specimen were used for computation of sample means, which were in turn used for calculation of site means. Paleomagnetic software for an IBM PC, by Randy Enkin and Stanislav Shipunov, and for a Macintosh, by Jean-Pascal Cogné, were used for analysis.

4. Results

4.1. Zaalay Range (locality 1)

Most samples of the Late Triassic–Early Jurassic Minteke Formation and green varieties of the Upper Permian Gundara Formation showed erratic behaviour during the cleaning and were discarded. In contrast, violet and red varieties from the same formation responded well to thermal demagnetization. Apart from an unstable remanence removed after the first heating, a single characteristic component, ChRM, was isolated from many samples (Fig. 2a). In other samples, however, two components were recognized: the first one with unblocking temperatures not exceeding 600°C, ITC, and the ChRM at higher temperatures (Fig. 2b). ChRM site means are better grouped after tilt correction than in situ (Fig. 3a,b), and the fold test [21] for the ChRM is positive (Table 1). For the ITC data, the fold test is inconclusive. The overall mean of the ChRM is slightly steeper than that of the ITC, but this difference is statistically insignificant (Table 1). The origin of the ITC remained unclear, and this component was excluded from further analysis.

In Upper Permian conglomerates, a well-defined ChRM showing rectilinear decay to the origin was isolated from 16 boulders (Fig. 2c) out of 20 studied. The length of vector resultant is 0.22 for 16 ChRM directions, whereas the 95% critical value is 0.40 [22]. Thus the conglomerate test is positive (Fig. 3c).

It is not always clear whether the boulders are from the Gundara Formation, and, therefore, the conglomerate test does not relate directly to the Late Permian result. The ChRMs in the Gundara red beds and boulders from overlying conglomerates, however, often have similar unblocking temperatures (Fig. 2a–c). Since it does not seem likely that the Gundara red beds had been completely remagnetized, while the boulders in overlying conglomerates retained their pre-depositional magnetization, the ChRM in the Gundara red beds pre-dates the folding and accumulation of conglomerates and is most probably primary. The reversed polarity of the Gundara rocks also agrees with the inferred Permian age of the ChRM.

4.2. Peter the First Range (locality 2)

A northerly and downward remanence, often accounting for more than 70% of the NRM at site S1

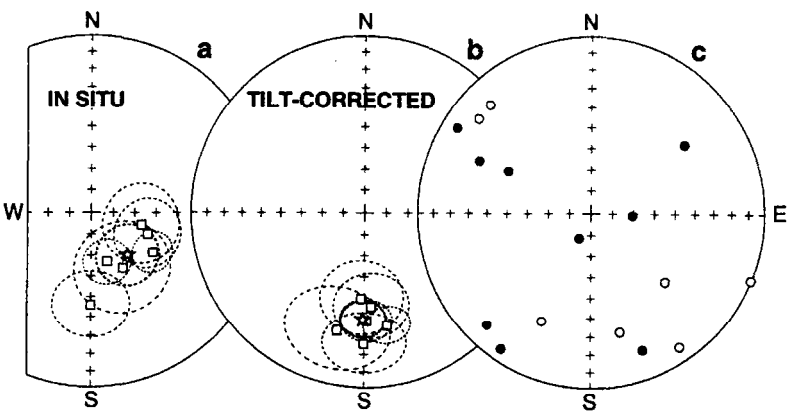


Fig. 3. Equal-area projection of ChRM site means (□) together with their confidence circles (thin dashed lines) and the overall mean (star) with its confidence circle (thick dashed line) from the Upper Permian Gundara formation at locality 1 (a) in situ and (b) after tilt correction. (c) Tilt-corrected ChRM directions in pebbles from Upper Permian conglomerates. □,○ and star = projected onto upper hemisphere; ● = projected onto lower hemisphere.

Table 1
Paleomagnetic data from the Zaalay Range (locality 1; 38.2°N, 72.3°E)

S	N	B	In situ				Tilt-corrected			
			D°	I°	k	a95°	D°	I°	k	a95°
Red beds of the Gundara Formation										
Intermediate-temperature component										
K2	11/4	15/30	157.8	−52.8	15	24.7	170.9	−26.7	15	24.7
K5	5/4	37/45	138.9	−43.1	10	30.8	166.6	−22.4	11	29.4
Mean (samples) [a]	34/15		154.2	−56.9	18	8.8	174.3	−32.0	20	8.3
High-temperature component										
K1	4/3	0/20	180.1	−45.8	56	10.8	180.1	−25.8	56	10.8
K2	11/10	15/30	162.5	−65.6	21	9.7	178.7	−37.9	21	9.7
K3	5/4	48/46	150.8	−59.9	19	16.3	193.1	−31.4	19	16.3
K4	5/4	30/45	123.4	−56.9	91	7.4	168.5	−34.7	91	7.4
K5	5/3	37/45	103.8	−66.2	48	11.7	182.0	−49.3	47	11.8
K6	6/5	36/45	111.0	−62.0	25	12.6	175.9	−45.2	23	13.1
Mean (samples)	34/29		143.7	−63.5	17	6.4	179.3	−38.0	23	5.4
$F_{(10,46)} = 2.04$			f = 4.28				f = 1.69			
Mean (sites)	6		141.8	−62.5	23	12.0	179.8	−37.6	55	7.7
Upper Permian conglomerates										
Intermediate-temperature component										
Y1	20/6	30/45	196.1	−36.9	–	–	198.8	7.2	8.3	24.7

[a] At the other four sites this component was isolated from just one or two samples. S = the number of site; N = the number of samples studied/accepted; B = dip direction/dip; D = declination; I = inclination; K = concentration parameter; a95 = the radii of confidence circle; F = the critical value of F statistics with the number of degrees of freedom in brackets; f = calculated values of the same statistics [21].

(Fig. 2d) and much weaker at site S2 (Fig. 2f,g), was removed at 200°C from most samples of the Gundara Formation. Its in situ mean direction is close to that of the present-day field (INI in Table 2). Apart from this unstable remanence, a single steep-up and eastward component (HTC) persisting up to 660°C was isolated from sandstones at site S1 (Fig. 2d,e; Table 2). The corresponding linear segments usually miss the origin, thus inferring the presence of still another component. Its isolation or reliable definition of remagnetization great circles, however, was prevented by the erratic directional behaviour of most samples above 600°C.

A single component showing rectilinear decay to the origin was isolated above 200°C from some samples at site S2 (Fig. 2d); in other samples, how-

Table 2
Paleomagnetic data from the Peter the First Range (locality 2; 38.8°N, 71.2°E)

S	C	N	B	In situ				Tilt-corrected			
				D°	I°	k	a95°	D°	I°	k	a95°
S1	INI	14/14	5/45	11.3	36.6	13	8.1	10.1	−8.2	13	8.1
S1	HTC	14/14	5/45	37.7	−36.0	26	7.9	86.4	−63.7	26	7.9
S2	INI	23/20	355/60	331.7	54.9	17	7.8	341.8	−2.8	17	7.8
S2	LTC	23/10	355/60	3.9	33.1	14	13.5	3.4	−26.3	14	13.5
S2	ITC	23/15	355/60	34.4	−48.4	15	10.1	127.8	−54.9	15	10.1
S2	HTC	23/20	355/60	52.8	−57.6	26	6.6	137.4	−42.0	26	6.6
INI mean for the locality (samples)											
	INI	37/34		348.9	50.3	10	5.8	352.0	−3.8	10	5.7

C = paleomagnetic component; INI = remanence destroyed after the first heating to 200°; LTC, ITC and HTC = low-, intermediate- and high-temperature component, respectively. Other notation as in Table 1.

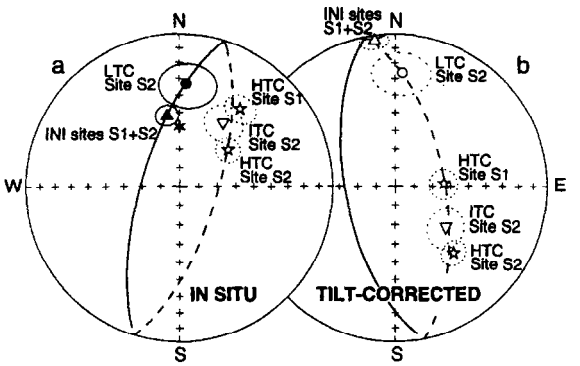


Fig. 4. Equal-area projection of mean directions of different components isolated from the Upper Permian Gundara Formation at locality 2 with associated confidence circles (a) in situ and (b) after tilt correction. The thick line is the best-fitting plane adjusted to all mean directions. Asterisk is the present-day dipole field at the locality. Open symbols and dashed lines are projected onto upper hemisphere; solid symbols and lines are projected onto the lower hemisphere.

ever, up to three linear segments were identified (Fig. 2g). Arbitrarily, a lower temperature component, LTC, with unblocking temperatures below 400°C and an intermediate component, ITC, with unblocking temperatures within the magnetite range (above 400°C and up to 600°C) were recognized. After removal of the LTC and ITC, a high-temperature component (HTC) was also isolated from most samples.

Mean HTC directions from sites S1 and S2 are different in both coordinates (Table 2; Fig. 4). Mean directions of all isolated components lie on a single great circle stretching from the INI to the HTC mean for site S2 (Fig. 4). It is likely that only two components are actually present in these rocks; namely, a magnetization of normal polarity with the direction close to the present-day field and a reversed component. All other 'components' are the result of partial (LTC and ITC at site S2) or complete (HTC at site S1) overlapping of unblocking spectra of these true components. The best approximation to the first component is given by the overall INI mean (Table 2); judging by the proximity of the in situ INI mean to the present day field, this remanence is post folding. The true direction of the reversed compo-

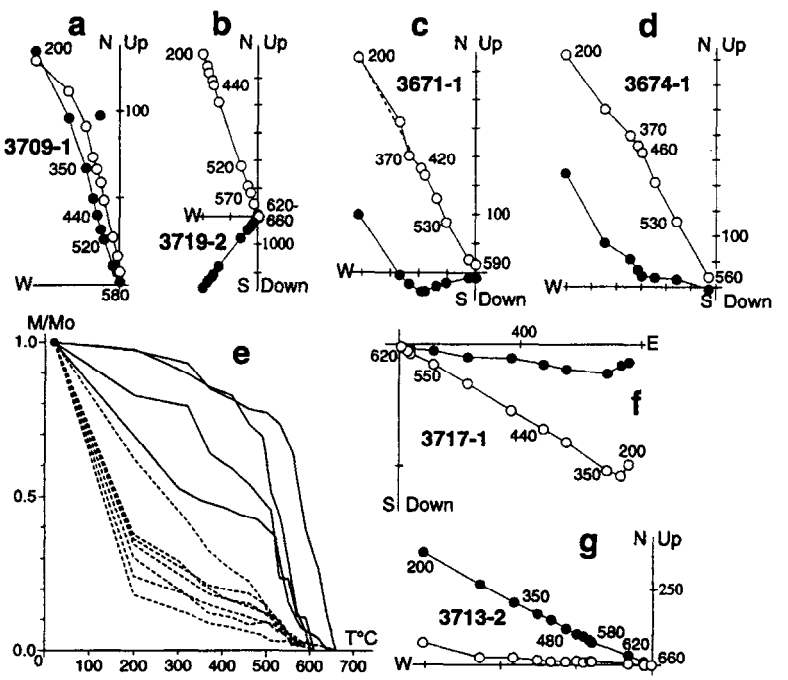


Fig. 5. Representative thermal demagnetization plots (a–d and f,g) and normalized intensity curves (e) of samples from Triassic volcanics (locality 3). (a–d) 'Stable' (3671, 3719) and 'unstable' (3674, 3709) samples from the main population of data from volcanics; (f) sample with anomalous flat directions; (g) lava boulder (see text for explanation). Solid lines in (e) denote 'stable' and dashed lines 'unstable' samples. Other notation as in Fig. 2. NRM value on the plots is omitted for clarity.

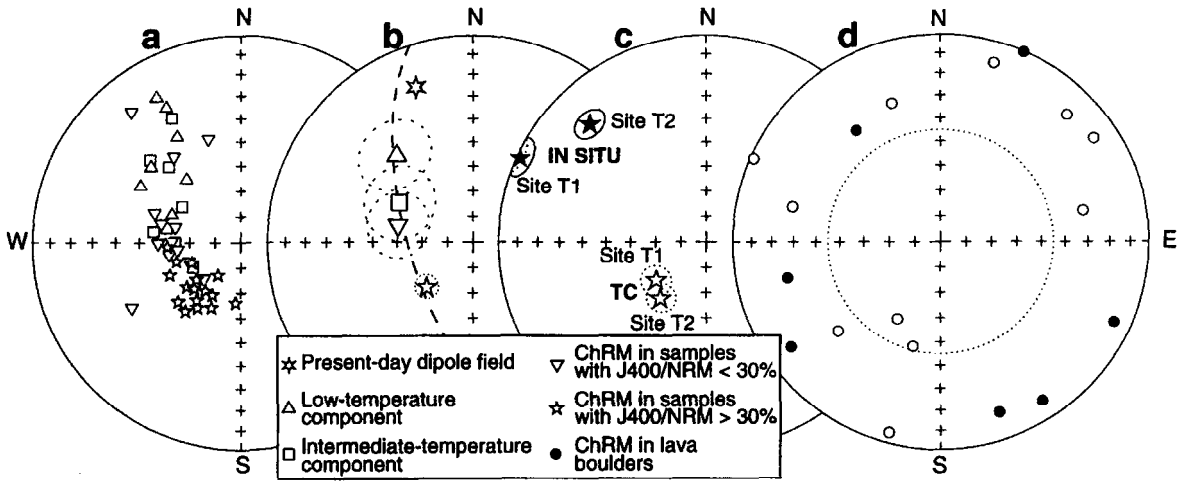


Fig. 6. (a) Unit vectors different components isolated from Triassic volcanics at locality 3. (b) Mean directions of the same components with associated confidence circles; thick line is the best-fitting plane adjusted to all mean directions. (c) In situ and tilt-corrected (TC) ChRM mean directions of 'stable' samples from sites T1 and T2. (d) ChRM directions from lava boulders; dotted line shows the area almost without paleomagnetic directions. All directions in (a), (b) and (c) are after tilt correction. Open symbols and dashed lines projected onto upper hemisphere; solid symbols and lines projected onto lower hemisphere.

nent is best approximated by the HTC mean from site S2. The fold test for latter result, however, is inconclusive.

4.3. Darvaz Range (locality 3)

The NRM intensities in the Triassic volcanics drop a between few percent to more than four-fold at 200°C; intensity decay at higher temperatures varies too. Some samples reveal rectilinear decay to the origin above 200° or 300°C (Fig. 5a,b), whereas additional rectilinear segments can be recognized on other plots at low to intermediate temperatures (Fig. 5c,d). The isolated 'components' were combined into the low-temperature (from 200°C to 400°C) and intermediate-temperature groups (from 300°C to 480°C

Table 3
Paleomagnetic results on Triassic volcanics of the Darvaz Range (locality 3; 38.7°N, 71.0°E)

S	N	B	In situ				Tilt-corrected			
			D°	I°	k	a95°	D°	I°	k	a95°
Low-temperature component (from 200° to 350° or 400°)										
T1	30/8	321/97	308.8	56.0	7	18.5	317.3	−37.1	12	14.2
Intermediate-temperature component (from 300° to 450° or 500°)										
T1	30/9	323/93	306.2	31.4	19	10.8	297.0	−57.6	18	11.0
High-temperature component, samples with J400/NRM < 30% [a]										
T1	30/13	322/89	303.1	21.8	11	11.8	283.0	−59.6	12	11.3
High-temperature component, samples with J400/NRM > 30%										
T1	30/10	322/89	294.7	2.3	56	5.9	232.1	−64.9	55	6.0
T2	13/6	251/129	315.5	21.0	100	5.7	218.9	−61.6	100	5.7
Mean (samples)	43/16		302.2	9.5	24	7.2	226.8	−63.8	63	4.4
$F_{(2,28)} = 3.34$			f = 27.77				f = 1.68			
Mean (samples J400/NRM > 50%)	43/11		303.7	10.7	24	8.6	223.8	−65.6	84	4.6
Flat directions										
T1,T2	43/6		52.8	−29.9	23	14.3	88.5	13.1	13	19.0

[a] Only one sample from section 2 falls into this category. Notation as in Table 1.

or 520°C); finally, the ChRM directions were pooled irrespective of temperature interval. Apart from a few 'anomalous' samples (see below), the directions of all components are confined to a band on the upper hemisphere; moreover, the 'ChRM' directions are also in a band (Fig. 6a). The 'component' mean vectors lie on a great circle which passes close to the present-day dipole field direction (Fig. 6b).

The ChRM direction in a sample does not correlate with the temperature interval where it is isolated. As NRM intensity decay after the first cleaning steps varies greatly from sample to sample (Fig. 5e), the ratio of magnetization intensity at 400°C to its initial value (J400/NRM) was chosen as the discrimination parameter. At this temperature, two subsets can be recognized, with J400/NRM < 30% ('unstable') and J400/NRM > 30% ('stable') (Fig. 5e). Mean ChRM directions for these subsets fall on the same great circle (Fig. 6b), the 'stable' mean being the farthest from the present-day dipole field direction. A modification of the 'stable' subset by deleting specimens with the ratio above 40% and then 50% does not change the mean direction (Table 3). The fold test [21] points to a pre-folding age of the ChRM in the 'stable' subset (Fig. 6c), which includes samples from at least 12 volcanic units. Most samples from the 'unstable' subset and most samples where 'low'- and 'intermediate'-temperature components are isolated are from the vertically dipping section (site T1), and the fold test for them is inconclusive.

It is most likely that only a component aligned along the present-day dipole field and the southwestern and up component are actually present here. The planar distribution of unit vectors, perfect alignment of the mean directions along the great circle and rectilinear decay to the origin irrespective of the ChRM direction result from strongly overlapping unblocking spectra of these two components.

In six hand samples, another component was identified. This shallow inclination component (Fig. 5f) is quite different from the main body of data both in situ and after tilt-correction (Table 3). These six directions are better grouped in situ than after tilt correction; the fold test [23], however, is inconclusive.

A ChRM was isolated from 200°C or 300°C from all lava boulders (Fig. 5g). Its direction in sister specimens differs by a few degrees, whereas boulder means are apparently chaotic, and the calculated statistics of 0.151 are much less than the critical value of 0.377 of the Raleigh uniformity test [22]. A closer inspection, however, hints at some deviation from uniformity, since the distribution of boulder vectors is of girdle type (Fig. 6d). Despite this, the conglomerate test can be considered positive.

The anisotropy of magnetic susceptibility, AMS, is less than 3% in all samples, and the anisotropy of the ellipsoid axes shows no preferred orientation (Fig. 7) in either basalts, with steep or shallow ChRMs, or lava boulders. Thus AMS cannot account for the anomalous flat directions in volcanics and the girdle distribution in conglomerates.

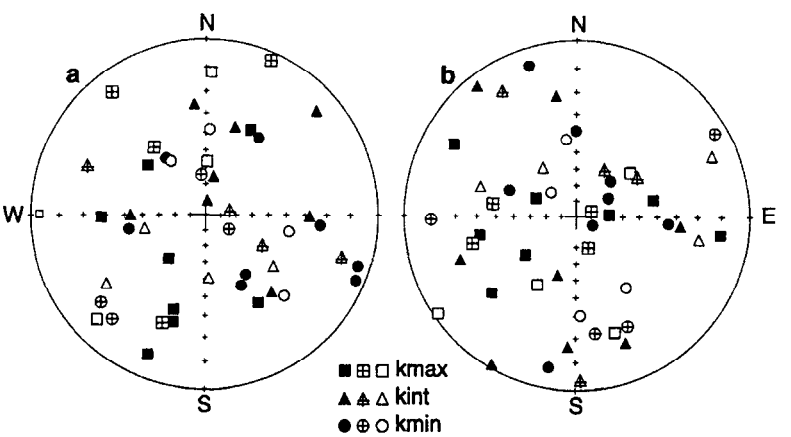


Fig. 7. Anisotropy ellipsoid axes from lava samples with steep (solid symbols) and flat (crossed symbols) ChRM directions and lava boulders (open symbols) (a) in situ and (b) after tilt correction. All symbols are on the lower hemisphere.

The reversed steep component in basalts is pre-folding and most probably primary, judging by the positive fold and conglomerate tests. However, the origin and significance of the girdle distribution of ChRM directions in lava boulders and flat component in volcanics remain unaccounted for, and the flat ChRMs were discarded.

5. Interpretation and discussion

The Permian result from locality 1 is confirmed by the positive conglomerate and fold tests. Unfortunately no field test could be performed on the Late Permian data from locality 2. However, these two mean inclinations are statistically the same (Tables 1 and 2). A rotation around vertical or inclined axis does not change the inclination of a pre-folding remanence [24,25]; hence, at least the necessary condition for the site S2 mean direction being of pre-folding origin is fulfilled.

The curvatures of the North Pamir and External zones are very similar (Fig. 1b). Cretaceous and Paleogene paleomagnetic results from the External Pamir zone imply that the curvature of this zone resulted from oroclinal bending of an initially almost rectilinear structure [9]. Both strike trends and mean declinations for localities 1 and 2 differ by about 45°, thus inferring the oroclinal bending of the North Pamir zone too. At the same time, the unbending of the Pamir brings the two tilt-corrected Permian directions into agreement and may be regarded as a positive fold test.

In principle, the Pamir data may be compared with the APWP for either the Indian, Siberian or European plates. Since this area was strongly affected by Alpine deformation, it seems to be reasonable to rely mainly on inclinations. The observed Permian inclinations are incompatible with the Indian reference data predicting a southern hemisphere position for the North Pamir, which is clearly not the case. The Siberia Permian poles are unreliable, not being based on adequate demagnetization and component analysis. The APWP for the East European plate is much better defined [26], but its validity for the Pamir is not straightforward because these two units are far away and separated by the fold belts of the Urals, Kazakhstan and Tien Shan, where considerable Permian horizontal movements have been proposed [27,28]. However, Permian paleomagnetic results from north-central Kazakhstan [29] agree well with the European reference data. Further southeastward and closer to the Pamir, Permian declinations over the Tien Shan strongly deflect with respect to the reference ones, but the inclinations fit quite well [30,31], implying no large latitudinal displacements between East Europe and Central Asia. Thus, comparison of the above results with the European APWP may reveal the movements of the North Pamir zone with respect to Eurasia.

The paleomagnetic result on Lower Cretaceous red beds from locality 3 is based on end-points after heating at 400°C [9]. A single characteristic component, however, was isolated from a few samples subjected to leaching in hydrochloride acid; the direction of this component is very close to that after thermal cleaning. Moreover, both the rocks and their demagnetization properties here mirror those from the South Darvaz Range, where the results after heating at 400°C and the main component isolated after complete demagnetization are in perfect agreement [8]. This means that the Cretaceous result from locality 3 may be used for further analysis. This remanence at the South Darvaz and elsewhere in the Tadjik basin further to the west, however, is thought to have been acquired during the Cretaceous Quiet Interval and is thus younger than the rocks. The same is probably true for locality 3 too, judging by the strong prevalence of normal polarity through the entire Lower Cretaceous section.

The observed Permian, Triassic and Cretaceous results agree well with the European reference data (Fig. 8), as well as the Permian inclination of 39° from the South Darvaz Range referred to by Klishevich et al. [11]. Moreover, the Pamir northward movement is thought to have taken place mainly in the Cenozoic [1–3]. The observed Cretaceous inclination overlaps both the European APWP [26] and the combined APWP for the last 200 Ma [32]. This pattern implies that the northward displacement of the North Pamir zone with respect to the East European plate since the Late Permian is within the error limits of the available data. One may hypothesize that the North Pamir moved at a large angle to reference directions, thus disguising the true magnitude of displacement. It is partly true for the Permian

and Triassic data, but the angle between the paleomeridians and displacement was hardly large enough to affect the interpretation. Moreover, this angle is negligible for the Cretaceous and later times.

Paleontological data show that the North Pamir zone belonged to the Eurasian fauna realm in the Late Permian [5]. This zone can be traced eastward into the Kunlun Range and westward into North Afghanistan [14,15]. Moreover, Carboniferous and Triassic volcanic belts of the North Pamir can be traced even further westward into the Kopet Dagh Range and, then, the North Caucasus [16]. It is difficult to conceive how these long and narrow structures could move over a large distance as continuous units. It is more likely that these belts marked the southern boundary of Eurasia and were later deformed. Using the Paleogene and Cretaceous declinations from the External Pamir zone, and assuming the continuity of the sedimentation basins, the Alpine northward displacement of the northern rim of the Pamir has been estimated to be about 300 km [9], which agrees with the observed Permo–Triassic inclinations.

The North Pamir zone and its counterparts to the west and east lie to the south of the Karakum block, in the west, and Tarim platform, in the east. If the above reasoning is correct it imposes strong constraints on the positions of these blocks, as illustrated in a schematic reconstruction (Fig. 9). Of course, this reconstruction is very preliminary, and should be further constrained with new paleomagnetic data.

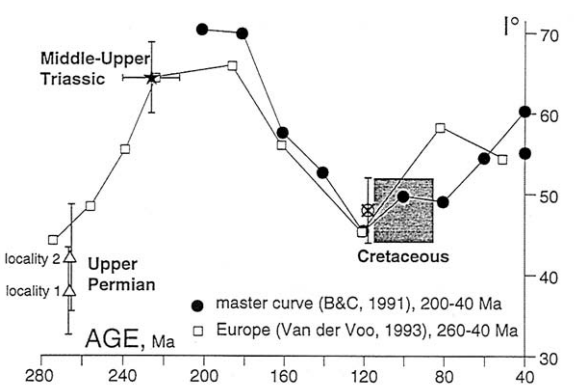


Fig. 8. Plot of inclination versus age for the Late Permian (triangles), Middle–Late Triassic (star) and Early Cretaceous (crossed open dot) data from the North Pamir and reference data. The remanence in Lower Cretaceous red beds from locality 3 is inferred to have been acquired during the Cretaceous long Quiet Interval [8] and thus should fall into the shaded rectangle. Open squares and dots are the reference inclinations recalculated from the APWP for Europe [26] and the master curve from Besse and Courtillot [32], respectively. Error bars are shown as solid lines.

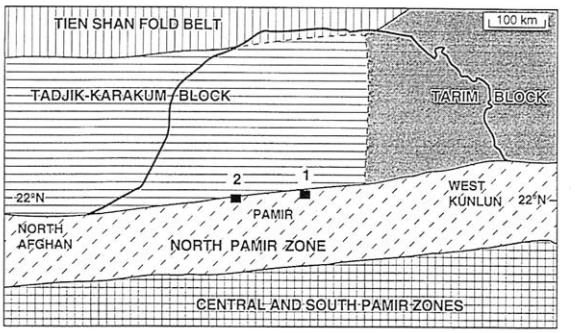


Fig. 9. Schematic reconstruction of the Pamir (same area as in Fig. 1b). The boundaries of the Tarim and Tadjik–Karakum blocks and Tien Shan fold belt now under the Pamir are absolutely arbitrary (thin dashed lines); the present-day boundary of the Pamir is shown as a thick solid line. The latitude of 22°N on the figure frame corresponds to the overall mean of the Permian data. Permian localities shown as solid rectangles are numbered as in the text and Fig. 1b.

6. Conclusions

Pre-folding, and most probably primary, components have been isolated from Upper Permian sediments and Middle–Upper Triassic volcanics of the North Pamir zone. The observed inclinations, together with the earlier results on Lower Cretaceous red beds, agree with the European reference data, thus implying the close proximity of the study area to Eurasia at least since the Late Permian. This, in turn, implies that the major part of the India–Eurasia convergence was consumed to the south of North Pamir. Paleomagnetic data from tectonic units situated between the studied area and the Indian platform are needed to evaluate the history of Central Asia from the Late Paleozoic through to the present.

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